U1: genuine Espressif ESP32-S3-DevKitC-1-N8 v1.1 (D-068), soldered on through its own headers. GPIO map v0.2 (PCB-10 README).

GPIO map v0.3 (layout): pins chosen so each track runs to its connector without crossings. Unused: GPIO0/3/45/46 (strapping), 19/20/43/44 (USB), 35/39/40 (spares), 38 (v1

J10-1: PB-SAFE-BASE 5 V (05-harness 5.1). D1 Schottky to the DevKitC 5V pin: a service USB cable can't back-feed W11 (D-068 E).

+3V3 is the DevKitC's own SGM2212-3.3 (800 mA); the carrier draws about 65 mA peak from it (lesson 1).

U3 TPS3436CCCBG (D-068 C): feed falling edge on WDI every 10 ms; valid window 5.5–18 ms, late fault <= 22 ms;

first edge within 500 ms of power-up; WDO low 200 ms resets the DevKitC (EN) and drops READY. MR pulled up (no PG reaches C3).

READY = WDO AND WD-EN AND MCU_RDY (BD-10, BC-08). MCU_RDY (GPIO2) pulled down: unpowered or

resetting reads not ready. 1k out to J10-12 (PCB-03 BASE_READY, fail-inactive).

J10-5/6 (power-boards 2.4): SLP from U4 through 1k each; IN1/IN2 LEDC with 100R; FLT wired-OR, 10k pull-up, 1k to GPIO9.

J10-2 (D-068 F, reordered at placement): 1 GND, 2/3 C0TX+/− (U2 A/B, 120R here), 4/5 C3TX−/+ (U2 Z/Y), 6 spare; U2's own pin order, so the pairs run straight. W25 1:1;

THVD1451: open/short/idle fail-safe, no bias needed. R18/R19 only with the MAX3490 alternate (open fail-safe only).

Encoders: MOT3001 Hall encoder on 3.3 V (vendor 3.3–5 V). 100R series; pull-ups DNP until the output type is known.

Rear TCRT (D-016): LED 100R from 3V3, switched low side by Q1 (GPIO47, 100k pull-down: off at boot).

TCRT_OUT 10k pull-up: unplugged or broken reads high = cliff = stop. 1k into GPIO8 (ADC1_CH7).

GP2Y_VO 10k/22k: 2.70 V max at 10 cm -> 1.86 V; a 5 V fault -> 3.44 V, under the S3's 3.6 V. 100n: the output updates every ~38 ms.

HALL_OUT: 1M to GND reads an open wire as 0 V (fault); 1k + 1n into GPIO10 (ADC1_CH9). PCB-15 has its own 1k/100n.

H1/H2: the two lower M2.5 screws (Z 86). The top edge sits in two printed hooks (D-068 A).

Spec: docs/03-build/04-pcbs/pcb-10-c3-carrier/README.md	
Espressif ESP32-S3-DevKitC-1 v1.1 carrier: TPS3436 watchdog, READY and sleep gates, RS-422 bas	
Makad	
Sheet: /	
File: pcb-10-c3-carrier.kicad_sch	
Title: PCB-10 C3 controller carrier	
Size: A2	Date: 2026-10-28
KiCad E.D.A. 10.99.0-5923-gde08734017	**Rev: A**
	Id: 1/1